

Shift Augmentation for Small-Image Digit Classification

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Abstract. We study whether one-pixel shift augmentation helps a multilayer perceptron on the 8x8 digits dataset. With identical hyperparameters, augmentation raises validation accuracy from 0.9722 to 0.9778 and macro F1 from 0.9722 to 0.9781.

1 Introduction

Small images leave little room for invariances to be learned from data. Classic remedies are translations, rotations and elastic distortions. Here we isolate the effect of translations on a fully connected network, which, unlike a convolutional one, has no built-in shift invariance.

2 Method

Every training image is copied four times and shifted by one pixel left, right, up and down, with zero padding. The training set therefore grows five times, from 1077 to 5385 images. Validation and test images are not augmented. The split is stratified 60/20/20 with seed 0.

The network has one hidden layer of 128 units, learning rate $3e-3$, L2 penalty $1e-4$, batch size 64 and at most 200 iterations; inputs are standardised.

3 Experiments

Table 1: Validation metrics with and without shift augmentation.

Training data	Accuracy	Macro F1	ROC-AUC
Original	0.9722	0.9722	0.9983
With shifts	0.9778	0.9781	0.9998

The gain is small but consistent across metrics. Most remaining errors are confusions between visually similar digits, as the confusion matrix in Figure 1 shows.

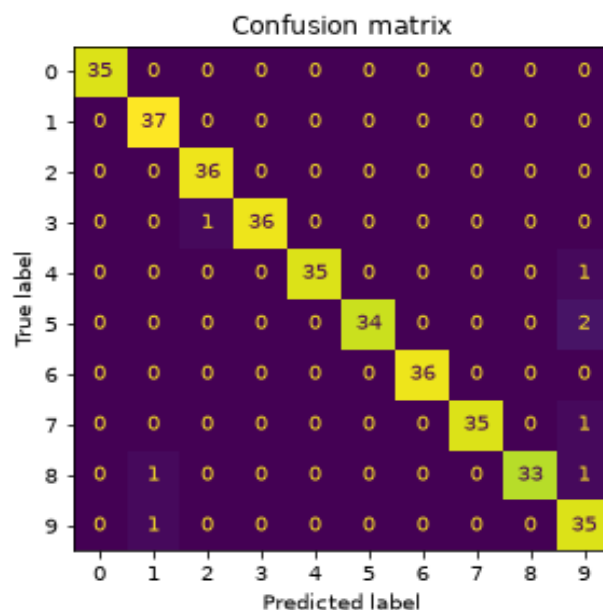


Figure 1: Confusion matrix of the augmented MLP on the validation set.

4 Conclusion

One-pixel shifts are a cheap augmentation that slightly improves an MLP on 8x8 digits. Rotations and a convolutional baseline are left for future work.